Results

Descriptives

Descriptives

	D_도입여부	소속	성별	역할	총공사비	나이	경력
N	0	39	39	39	39	39	39
	1	34	34	34	34	34	34
Missing	0	0	0	0	0	0	0
	1	0	0	0	0	0	0
Mean	0	1.31	1.15	1.92	2.92	46.9	17.5
	1	1.50	1.00	1.85	2.91	46.4	14.2
Median	0	1	1	2	3	48	15
	1	1.50	1.00	2.00	3.00	47.0	10.5
Standard deviation	0	0.468	0.366	0.839	1.09	12.0	12.4
	1	0.508	0.00	0.892	0.900	11.1	10.8
Minimum	0	1	1	1	1	24	1
	1	1	1	1	1	30	1
Maximum	0	2	2	3	4	67	45
	1	2	1	3	4	66	37

Frequencies

Frequencies of 소속

소속	D_도입여부	Counts	% of Total	Cumulative %
1	0	27	37.0%	37.0%
	1	17	23.3%	60.3%
2	0	12	16.4%	76.7%
	1	17	23.3%	100.0%

Frequencies of 성별

성별	D_도입여부	Counts	% of Total	Cumulative %
1	0	33	45.2%	45.2%
	1	34	46.6%	91.8%
2	0	6	8.2%	100.0%
	1	0	0.0%	100.0%

Frequencies of 역할

역할	D_도입여부	Counts	% of Total	Cumulative %
1	0	15	20.5%	20.5%
	1	16	21.9%	42.5%
2	0	12	16.4%	58.9%
	1	7	9.6%	68.5%
3	0	12	16.4%	84.9%
	1	11	15.1%	100.0%

Frequencies of 총공사비

총공사비	D_도입여부	Counts	% of Total	Cumulative %
1	0	5	6.8%	6.8%
	1	3	4.1%	11.0%
2	0	9	12.3%	23.3%
	1	6	8.2%	31.5%
3	0	9	12.3%	43.8%
	1	16	21.9%	65.8%
4	0	16	21.9%	87.7%
	1	9	12.3%	100.0%

Contingency Tables

Contingency Tables

		D_도입여부		
A7_인지경로		0	1	Total
1	Observed	2	0	2
	% within column	5.1%	0.0%	2.7%
2	Observed	27	10	37
	% within column	69.2%	29.4%	50.7%
3	Observed	7	7	14
	% within column	17.9%	20.6%	19.2%
4	Observed	0	17	17
	% within column	0.0%	50.0%	23.3%
5	Observed	3	0	3
	% within column	7.7%	0.0%	4.1%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	29.6	4	<.001
Fisher's exact test			<.001
N	73		

Contingency Tables

Contingency Tables

		D_도입여부		
A8_인지시점		0	1	Total
1	Observed	5	6	11
	% within column	12.8%	17.6%	15.1%
2	Observed	29	10	39
	% within column	74.4%	29.4%	53.4%
3	Observed	2	15	17
	% within column	5.1%	44.1%	23.3%
4	Observed	3	3	6
	% within column	7.7%	8.8%	8.2%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	19.0	3	<.001
Fisher's exact test			<.001
N	73		

Contingency Tables

Contingency Tables

A9_전달시간		D_도입여부 0	D_도입여부 1	Total
1	Observed	15	11	26
	% within column	38.5%	32.4%	35.6%
2	Observed	15	18	33
	% within column	38.5%	52.9%	45.2%
3	Observed	7	4	11
	% within column	17.9%	11.8%	15.1%
4	Observed	2	1	3
	% within column	5.1%	2.9%	4.1%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	1.71	3	.636
Fisher's exact test			.663
N	73		

Reliability Analysis

Scale Reliability Statistics

	Cronbach's α
scale	0.937

[3]

Item Reliability Statistics

	Item-rest correlation	If item dropped Cronbach's α
B1	0.765	0.930
B2	0.746	0.932
B3	0.779	0.929
B4	0.811	0.926
B5	0.707	0.934
B6	0.776	0.930
B7	0.857	0.924
B8	0.844	0.925

Correlation Matrix

Correlation Matrix

		B3	B4
B3	Pearson's r	—	
	df	—	
	p-value	—	
B4	Pearson's r	0.721	—
	df	71	—
	p-value	<.001	—

Correlation Matrix

Correlation Matrix

		B5	B6
B5	Pearson's r	—	
	df	—	
	p-value	—	
B6	Pearson's r	0.631	—
	df	71	—
	p-value	<.001	—

Correlation Matrix

Correlation Matrix

		B7	B8
B7	Pearson's r	—	
	df	—	
	p-value	—	
B8	Pearson's r	0.741	—
	df	71	—
	p-value	<.001	—

Exploratory Factor Analysis

Factor Loadings

	Factor	
	1	Uniqueness
B1	0.793	0.371
B2	0.773	0.402
B3	0.807	0.348
B4	0.842	0.291
B5	0.727	0.471
B6	0.810	0.344
B7	0.888	0.212
B8	0.875	0.234

Note. 'Principal axis factoring' extraction method was used in combination with a 'oblimin' rotation

[3]

Factor Statistics

Summary

Factor	SS Loadings	% of Variance	Cumulative %
1	5.33	66.6	66.6

Assumption Checks

Bartlett's Test of Sphericity

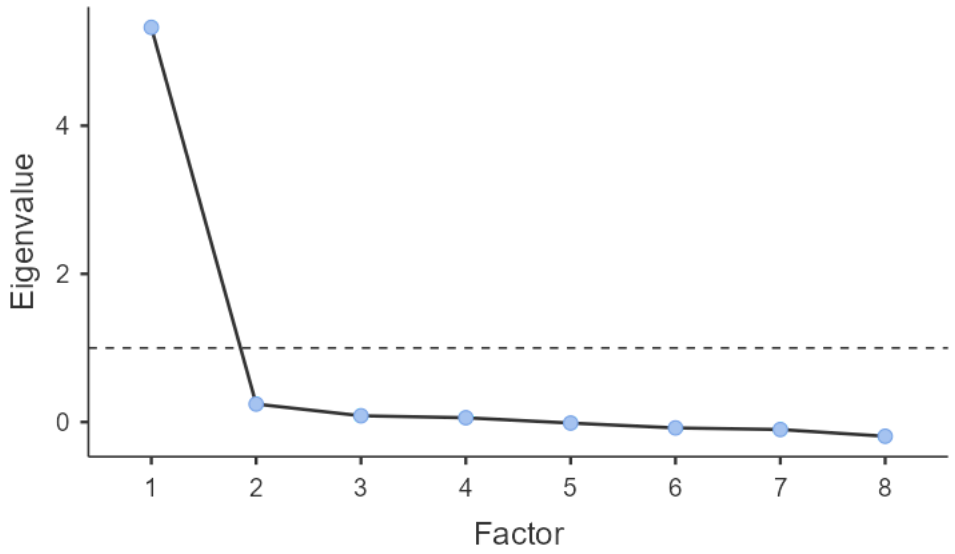
χ^2	df	p
456	28	<.001

KMO Measure of Sampling Adequacy

	MSA
Overall	0.911
B1	0.890
B2	0.945
B3	0.907
B4	0.920
B5	0.897
B6	0.892
B7	0.903
B8	0.933

Eigenvalues

Scree Plot



Descriptives

Descriptives

	D_도입여부	N	Missing	Mean	Median	SD	Minimum	Maximum	Skewness		Kurtosis	
									Skewness	SE	Kurtosis	SE
Y_실효성	0	39	0	3.48	3.50	0.812	1.38	5.00	-0.1462	0.378	0.349	0.741
	1	34	0	4.11	4.06	0.606	2.75	5.00	-0.2508	0.403	-0.322	0.788
Y1_조기인지	0	39	0	3.51	4	0.823	1	5	-0.6400	0.378	1.252	0.741
	1	34	0	4.18	4.00	0.758	2	5	-0.7563	0.403	0.639	0.788
Y2_대응신속성	0	39	0	3.87	4	0.833	2	5	-0.3255	0.378	-0.375	0.741
	1	34	0	4.24	4.00	0.699	3	5	-0.3603	0.403	-0.834	0.788
Y3_원격파악	0	39	0	3.32	3.50	0.997	1.00	5.00	-0.3139	0.378	-0.335	0.741
	1	34	0	4.12	4.00	0.708	2.50	5.00	-0.3078	0.403	-0.690	0.788
Y4_순찰부담경감	0	39	0	3.26	3.00	1.006	1.00	5.00	-0.0402	0.378	-0.355	0.741
	1	34	0	3.88	4.00	0.759	2.00	5.00	-0.2374	0.403	-0.148	0.788
Y5_위험통제력	0	39	0	3.67	3.50	0.814	1.50	5.00	-0.3417	0.378	0.385	0.741
	1	34	0	4.25	4.00	0.631	3.00	5.00	-0.2888	0.403	-0.649	0.788

Descriptives

Descriptives

	N	Missing	Mean	SD
C1	34	39	3.71	1.001
C2	34	39	3.71	0.970
C3	34	39	3.94	0.694
C4	34	39	3.53	0.992
C5	34	39	3.56	0.960
C6	34	39	3.47	0.961
C7	34	39	4.12	0.769
C8	34	39	3.85	0.857
D1	34	39	3.53	1.285
D2	34	39	3.59	1.209
D3	34	39	3.91	0.965
D4	34	39	3.59	1.131
Csub1_인프라범용성	34	39	3.71	0.880
Csub2_탐지제어안정성	34	39	3.74	0.720
Csub3_오탐필터링	34	39	3.51	0.909
Csub4_알림전파즉각성	34	39	3.99	0.744
C_시스템특성	34	39	3.74	0.663
D_경보피로완화	34	39	3.65	0.971

Independent Samples T-Test

Independent Samples T-Test

							95% Confidence Interval	
		Statistic	df	p		Effect Size	Lower	Upper
나이	Student's t	0.200	71.0	.842	Cohen's d	0.0469	-0.413	0.507
경력	Student's t	1.214	71.0	.229	Cohen's d	0.2848	-0.178	0.746

Note. H_a $\mu_0 \neq \mu_1$

Contingency Tables

Contingency Tables

소속		D_도입여부 0	D_도입여부 1	Total
1	Observed	27	17	44
	% within column	69.2%	50.0%	60.3%
2	Observed	12	17	29
	% within column	30.8%	50.0%	39.7%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	2.81	1	.094
Fisher's exact test			.150
N	73		

Nominal

	Value
Phi-coefficient	0.196
Cramer's V	0.196

Contingency Tables

Contingency Tables

성별		D_도입여부 0	D_도입여부 1	Total
1	Observed	33	34	67
	% within column	84.6%	100.0%	91.8%
2	Observed	6	0	6
	% within column	15.4%	0.0%	8.2%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	5.70	1	.017
Fisher's exact test			.027
N	73		

Nominal

	Value
Phi-coefficient	0.279
Cramer's V	0.279

Contingency Tables

Contingency Tables

		D_도입여부		
역할		0	1	Total
1	Observed	15	16	31
	% within column	38.5%	47.1%	42.5%
2	Observed	12	7	19
	% within column	30.8%	20.6%	26.0%
3	Observed	12	11	23
	% within column	30.8%	32.4%	31.5%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	1.05	2	.590
Fisher's exact test			.631
N	73		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.120

Contingency Tables

Contingency Tables

총공사비		D_도입여부 0	D_도입여부 1	Total
1	Observed	5	3	8
	% within column	12.8%	8.8%	11.0%
2	Observed	9	6	15
	% within column	23.1%	17.6%	20.5%
3	Observed	9	16	25
	% within column	23.1%	47.1%	34.2%
4	Observed	16	9	25
	% within column	41.0%	26.5%	34.2%
Total	Observed	39	34	73
	% within column	100.0%	100.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	4.70	3	.195
Fisher's exact test			.201
N	73		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.254

ANCOVA

ANCOVA - Y_실효성

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	4.1734	1	4.1734	8.5385	.005	0.121
소속	0.0581	1	0.0581	0.1188	.731	0.002
성별	1.1075	1	1.1075	2.2659	.137	0.035
역할	0.2419	2	0.1210	0.2475	.782	0.008
총공사비	1.6383	3	0.5461	1.1173	.349	0.051
나이	0.0176	1	0.0176	0.0360	.850	0.001
경력	0.0256	1	0.0256	0.0524	.820	0.001
Residuals	30.3038	62	0.4888			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

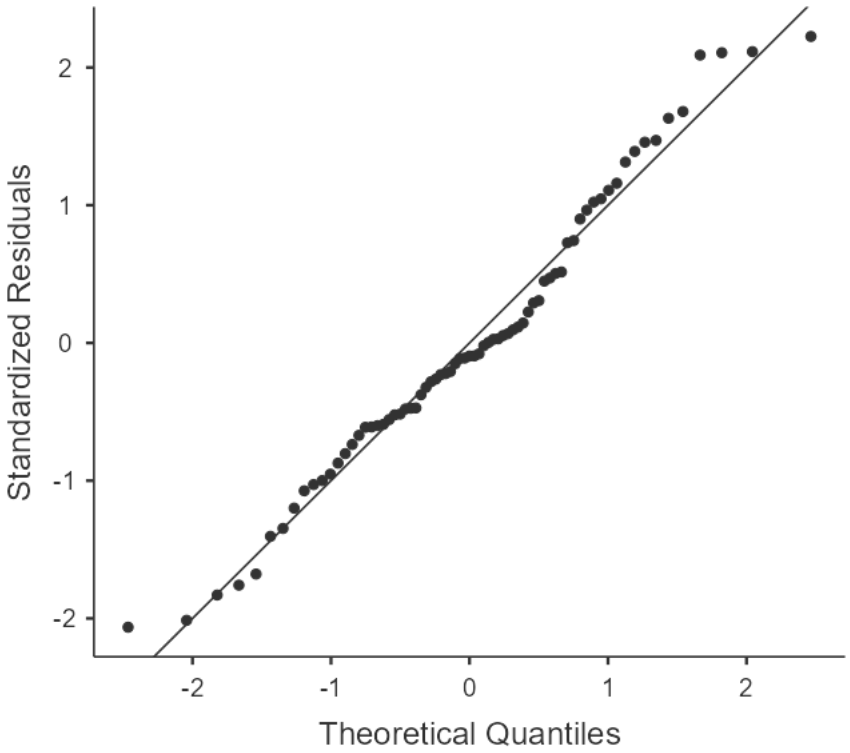
F	df1	df2	p
1.26	26	46	.245

[4]

Normality Test (Shapiro-Wilk)

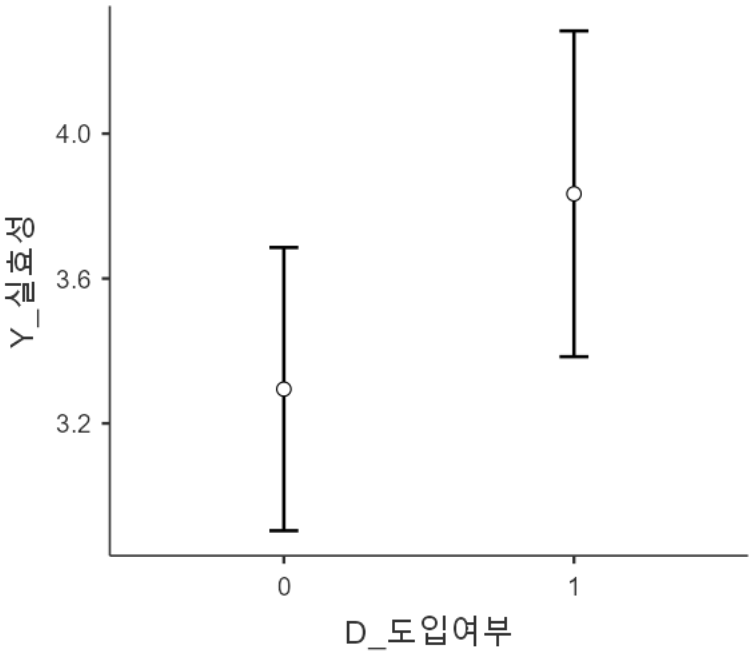
Statistic	p
0.974	.128

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.29	0.196	2.90	3.69
1	3.83	0.225	3.38	4.28

[5]

ANCOVA

ANCOVA - Y1_조기인지

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	5.577	1	5.577	10.500	.002	0.145
소속	0.116	1	0.116	0.219	.641	0.004
성별	1.401	1	1.401	2.638	.109	0.041
역할	0.417	2	0.209	0.393	.677	0.013
총공사비	2.969	3	0.990	1.863	.145	0.083
나이	1.088	1	1.088	2.049	.157	0.032
경력	1.356	1	1.356	2.553	.115	0.040
Residuals	32.932	62	0.531			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

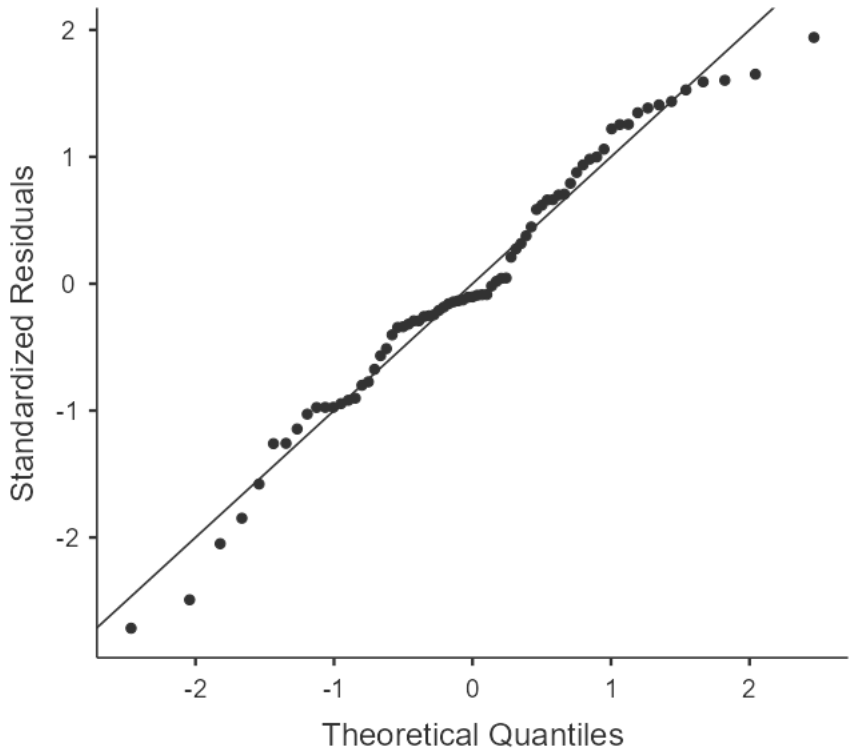
F	df1	df2	p
0.814	26	46	.709

[4]

Normality Test (Shapiro-Wilk)

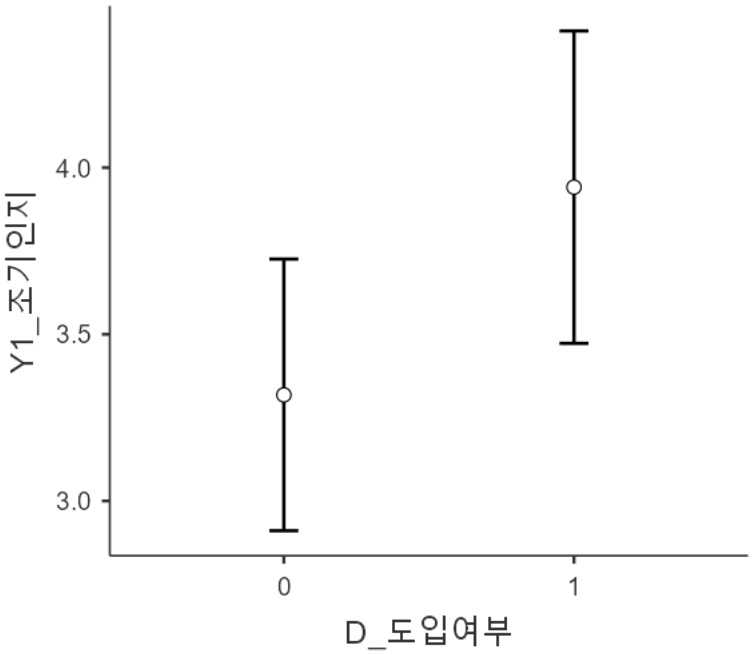
Statistic	p
0.975	.157

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.32	0.204	2.91	3.73
1	3.94	0.235	3.47	4.41

[5]

ANCOVA

ANCOVA - Y2_대응신속성

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	0.239	1	0.239	0.456	.502	0.007
소속	0.271	1	0.271	0.518	.474	0.008
성별	3.470	1	3.470	6.630	.012	0.097
역할	0.813	2	0.406	0.776	.464	0.024
총공사비	1.509	3	0.503	0.961	.417	0.044
나이	0.367	1	0.367	0.701	.406	0.011
경력	0.163	1	0.163	0.310	.579	0.005
Residuals	32.454	62	0.523			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

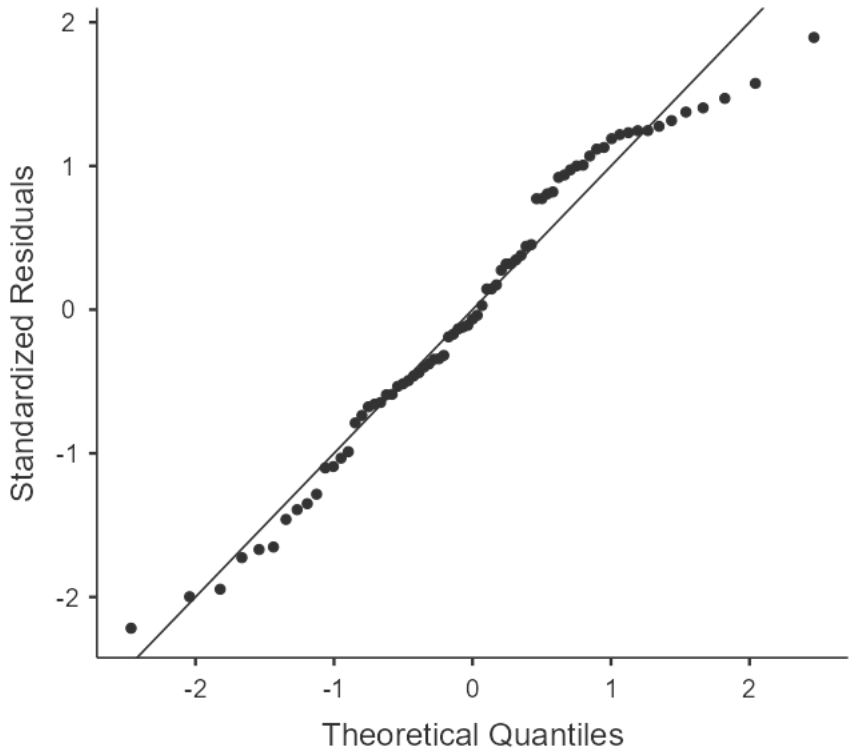
F	df1	df2	p
0.796	26	46	.731

[4]

Normality Test (Shapiro-Wilk)

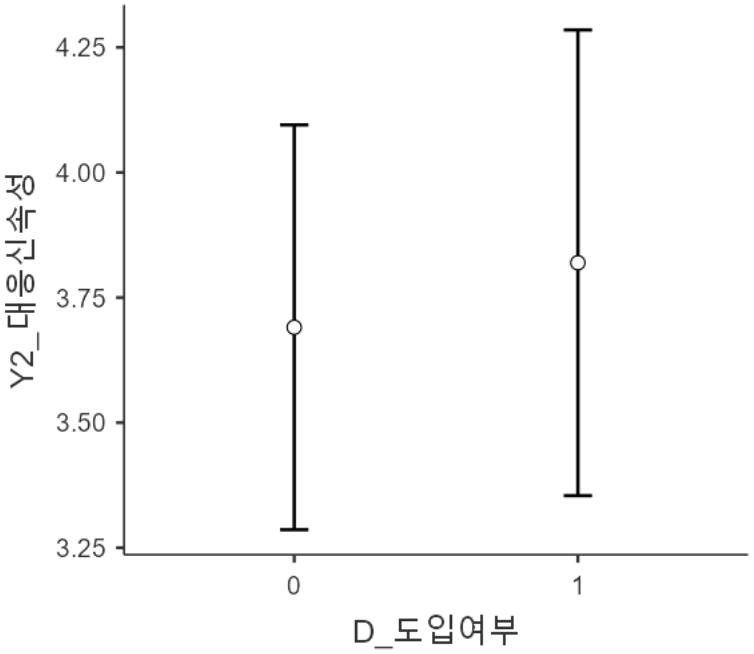
Statistic	p
0.969	.072

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.69	0.202	3.29	4.10
1	3.82	0.233	3.35	4.28

[5]

ANCOVA

ANCOVA - Y3_원격파악

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	7.4233	1	7.4233	9.3574	.003	0.131
소속	0.0683	1	0.0683	0.0861	.770	0.001
성별	0.6192	1	0.6192	0.7805	.380	0.012
역할	0.1833	2	0.0916	0.1155	.891	0.004
총공사비	2.0001	3	0.6667	0.8404	.477	0.039
나이	0.4113	1	0.4113	0.5185	.474	0.008
경력	0.1741	1	0.1741	0.2194	.641	0.004
Residuals	49.1852	62	0.7933			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

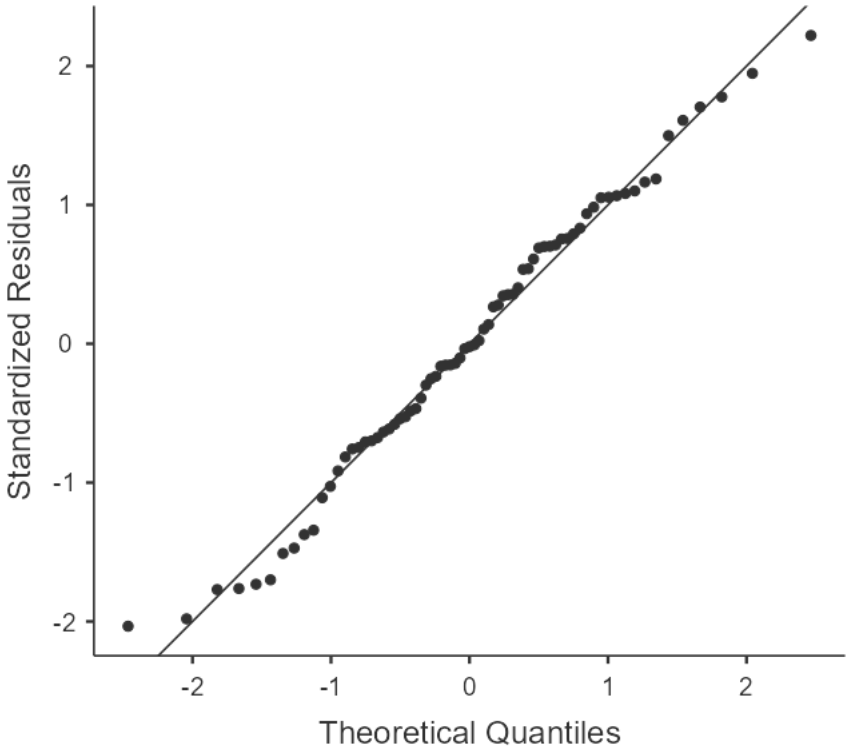
F	df1	df2	p
1.20	26	46	.289

[4]

Normality Test (Shapiro-Wilk)

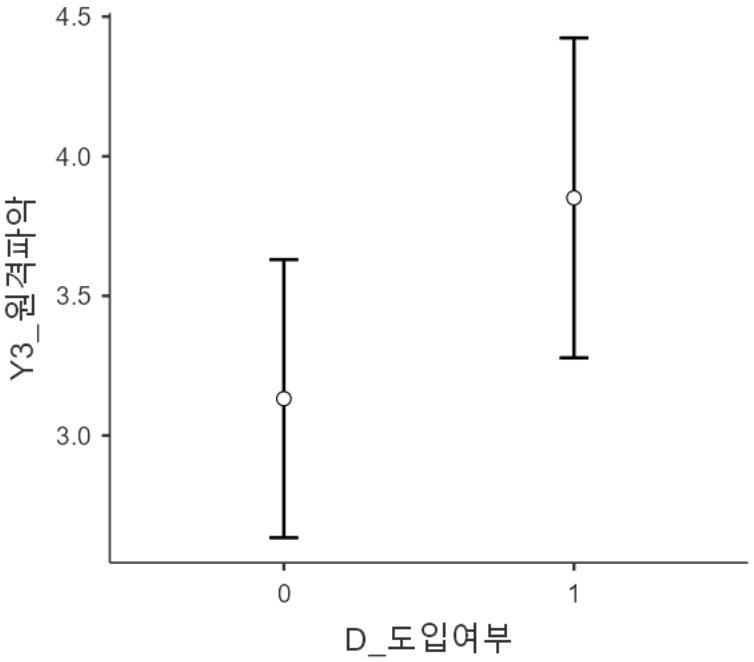
Statistic	p
0.984	.496

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.13	0.249	2.63	3.63
1	3.85	0.287	3.28	4.42

[5]

ANCOVA

ANCOVA - Y4_순찰부담경감

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	3.8790	1	3.8790	4.8607	.031	0.073
소속	0.0264	1	0.0264	0.0331	.856	0.001
성별	0.5815	1	0.5815	0.7287	.397	0.012
역할	0.4497	2	0.2248	0.2818	.755	0.009
총공사비	1.6115	3	0.5372	0.6731	.572	0.032
나이	0.1636	1	0.1636	0.2049	.652	0.003
경력	0.1196	1	0.1196	0.1499	.700	0.002
Residuals	49.4780	62	0.7980			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

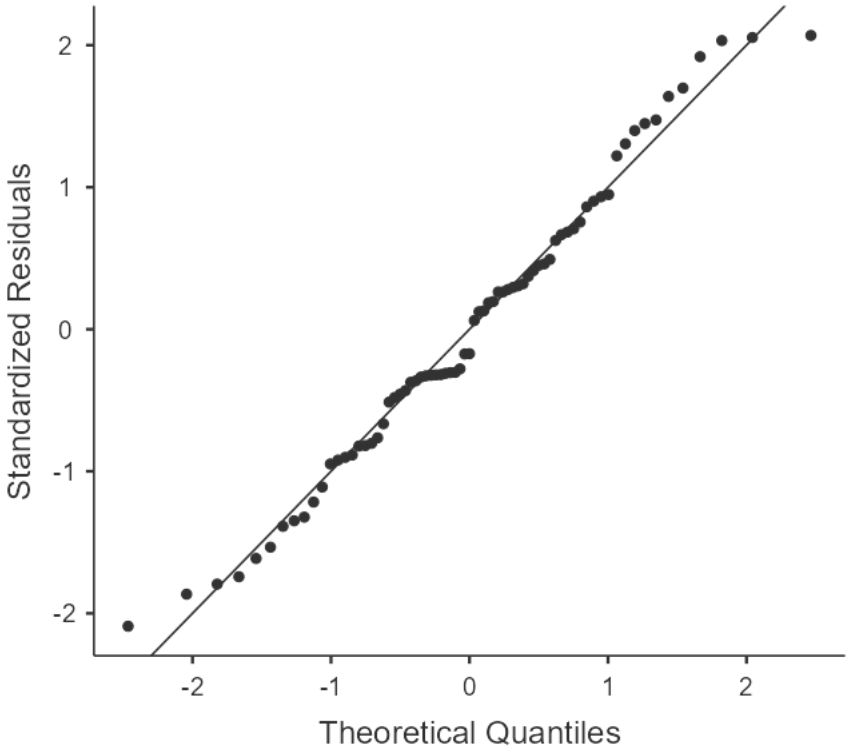
F	df1	df2	p
1.18	26	46	.304

[4]

Normality Test (Shapiro-Wilk)

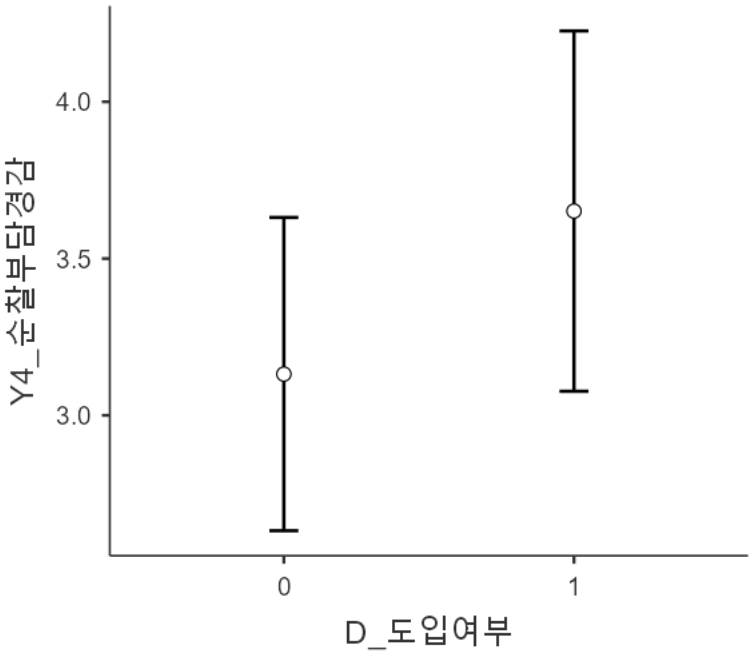
Statistic	p
0.982	.402

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.13	0.250	2.63	3.63
1	3.65	0.287	3.08	4.23

[5]

ANCOVA

ANCOVA - Y5_위험통제력

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	4.21184	1	4.21184	8.7446	.004	0.124
소속	0.39684	1	0.39684	0.8239	.368	0.013
성별	1.29223	1	1.29223	2.6829	.106	0.041
역할	0.13182	2	0.06591	0.1368	.872	0.004
총공사비	1.74361	3	0.58120	1.2067	.315	0.055
나이	0.00567	1	0.00567	0.0118	.914	0.000
경력	0.03546	1	0.03546	0.0736	.787	0.001
Residuals	29.86230	62	0.48165			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

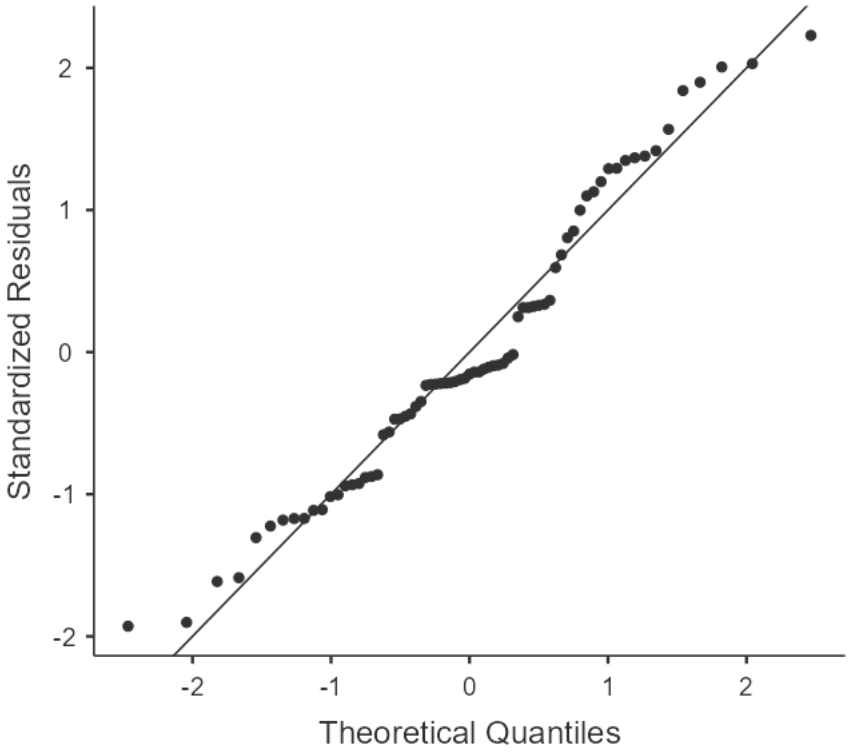
F	df1	df2	p
0.954	26	46	.541

[4]

Normality Test (Shapiro-Wilk)

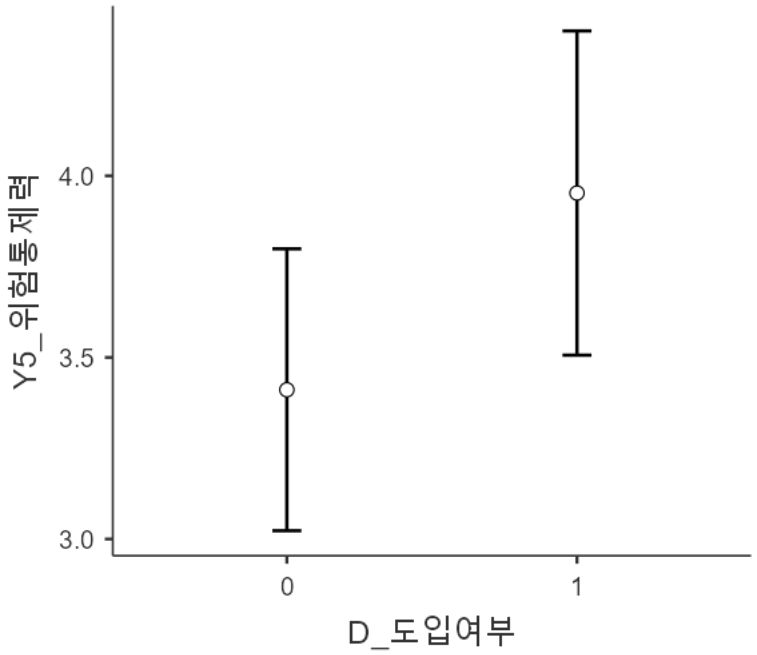
Statistic	p
0.964	.036

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.41	0.194	3.02	3.80
1	3.95	0.223	3.51	4.40

[5]

ANCOVA

ANCOVA - Y_실효성

	Sum of Squares	df	Mean Square	F	p	η^2p
D_도입여부	0.1714	1	0.1714	0.3481	.557	0.006
소속	0.0402	1	0.0402	0.0816	.776	0.001
성별	1.5959	1	1.5959	3.2398	.077	0.051
역할	0.2406	2	0.1203	0.2442	.784	0.008
총공사비	2.0596	3	0.6865	1.3938	.253	0.065
나이	0.0718	1	0.0718	0.1457	.704	0.002
경력	0.1349	1	0.1349	0.2739	.603	0.005
D_도입여부 ✻ 나이	0.7368	1	0.7368	1.4958	.226	0.024
D_도입여부 ✻ 경력	0.2716	1	0.2716	0.5514	.461	0.009
Residuals	29.5545	60	0.4926			

[4]

Assumption Checks

Homogeneity of Variances Test (Levene's)

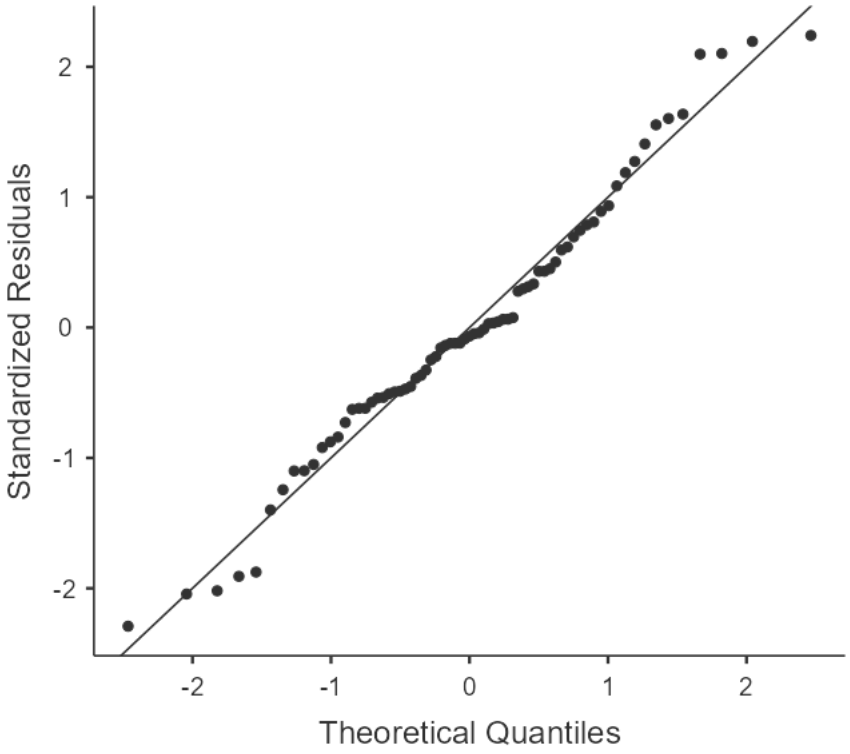
F	df1	df2	p
1.29	26	46	.222

[4]

Normality Test (Shapiro-Wilk)

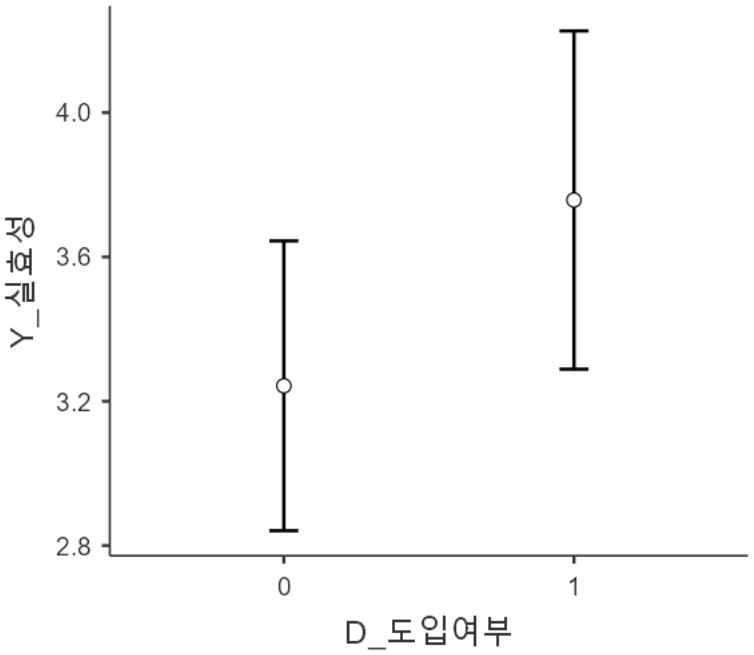
Statistic	p
0.976	.172

Q-Q Plot



Estimated Marginal Means

D_도입여부



Estimated Marginal Means - D_도입여부

			95% Confidence Interval	
D_도입여부	Mean	SE	Lower	Upper
0	3.24	0.201	2.84	3.64
1	3.76	0.234	3.29	4.23

[5]

Linear Regression

Model Fit Measures

Model	R	R^2
1	0.564	0.318

Note. Models estimated using sample size of N=73

Model Coefficients - Y_실효성

Predictor	Estimate	SE	95% Confidence Interval Lower	95% Confidence Interval Upper	t	p
Intercept[a]	3.32591	0.6636	1.9994	4.6524	5.0121	<.001
나이	-0.00218	0.0115	-0.0251	0.0208	-0.1898	.850
경력	0.00266	0.0116	-0.0205	0.0259	0.2290	.820
D_도입여부:						
1 – 0	0.53922	0.1845	0.1703	0.9081	2.9221	.005
소속:						
2 – 1	-0.13897	0.4031	-0.9448	0.6669	-0.3447	.731
성별:						
2 – 1	-0.52436	0.3483	-1.2207	0.1720	-1.5053	.137
역할:						
2 – 1	0.22011	0.3922	-0.5639	1.0041	0.5612	.577
3 – 1	0.30080	0.4284	-0.5556	1.1572	0.7021	.485
총공사비:						
2 – 1	0.35734	0.3185	-0.2794	0.9941	1.1218	.266
3 – 1	0.35749	0.3066	-0.2553	0.9703	1.1661	.248
4 – 1	0.02869	0.3148	-0.6006	0.6580	0.0911	.928

[a] Represents reference level

Data Summary

Cook's Distance

Mean	Median	SD	Range Min	Range Max
0.0172	0.00508	0.0266	6.29e-7	0.139

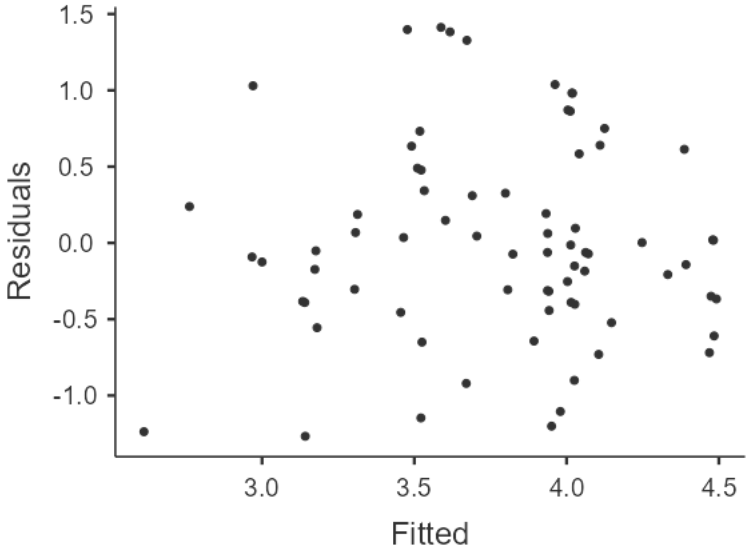
Assumption Checks

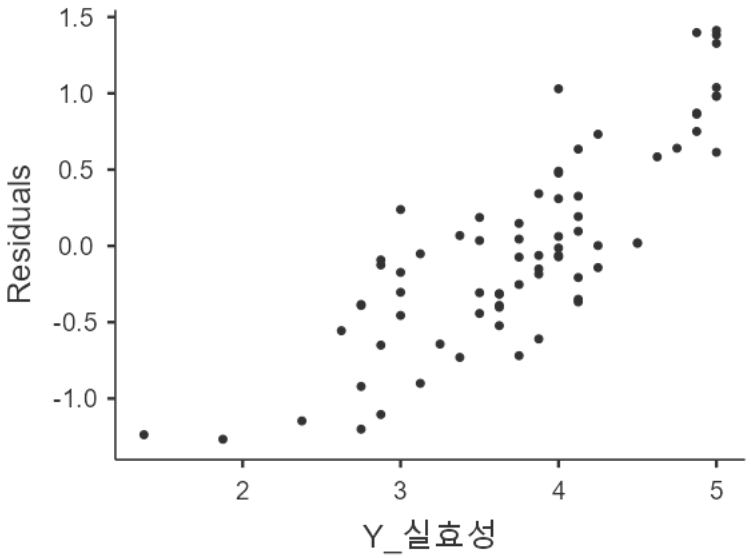
Collinearity Statistics

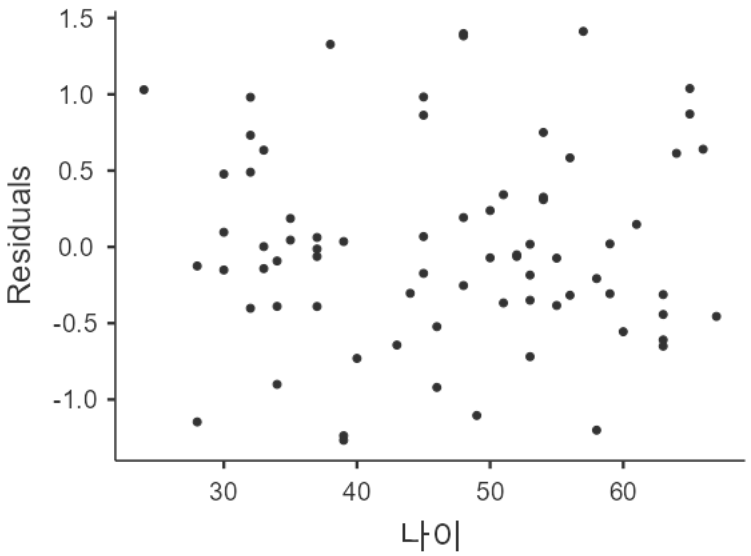
	VIF	Tolerance
나이	1.60	0.623
경력	1.65	0.605
D_도입여부	1.12	0.889
소속	2.41	0.415
성별	1.17	0.855
역할	1.75	0.572
총공사비	1.11	0.904

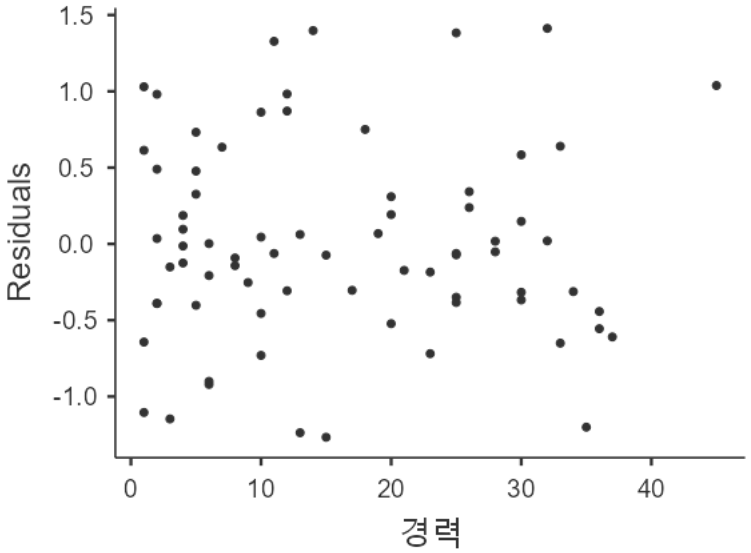
[4]

Residuals Plots









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[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).

[3] Revelle, W. (2025). *psych: Procedures for Psychological, Psychometric, and Personality Research*. [R package]. Retrieved from https://cran.r-project.org/package=psych.

[4] Fox, J., & Weisberg, S. (2024). *car: Companion to Applied Regression*. [R package]. Retrieved from https://cran.r-project.org/package=car.

[5] Lenth, R. (2025). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. [R package]. Retrieved from https://cran.r-project.org/package=emmeans.